

Priya, 41 · three months of morning stiffness > 1 hr · symmetric MCP, PIP, wrist, MTP swelling · positive RF and anti-CCP · early erosions on hand X-ray · **rheumatoid arthritis**

BIO 004 · WEEK 2 · CASE DEEP DIVE

Articulations and Joints

Morning stiffness that lasts two hours, swollen knuckles, and a synovium that has turned on itself.

PATIENT CASE

Priya is 41, a software engineer who has had **morning stiffness in her hands lasting more than an hour** for the past three months. She used to be able to open jars and play piano; now her hands feel "**locked**" until late morning.

On exam the **MCP and PIP joints of both hands** are warm, swollen, and tender, and the symptoms are **strikingly symmetric**. She also notes pain in **both wrists** and the **balls of both feet (MTP joints)**. She is fatigued. Labs come back with elevated CRP, **positive rheumatoid factor** and **positive anti-CCP antibodies**. Hand X-rays show **early periarticular erosions**.

RA is a disease of the synovial joint. To understand why Priya's small joints are being destroyed while her spine and DIPs are spared, you need to know what a synovial joint actually is, what holds it together, and what makes one go wrong.

GUIDING QUESTION

What are the three structural joint types, what is a synovial joint built from, and why does RA pick small synovial joints with symmetric distribution while sparing fibrous joints and the DIPs?

PREWORK

Companion to your Week 2 Joints prework page. Terms, video, and spaced recall for joint classification, fibrous, cartilaginous, synovial structure, accessory structures, movements, six types of synovial joints, factors affecting ROM, shoulder, knee, and common joint disorders live there.

Three ways joints can be built

Joints (articulations) are classified by what holds the bones together. There are three options:

● **Fibrous joints**

Bones held by dense connective tissue. No joint cavity. Little to no movement. Sutures of the skull, syndesmoses (tibiofibular), gomphoses (teeth in sockets).

● **Cartilaginous joints**

Bones held by cartilage. No joint cavity. Slight movement. *Synchondroses* (hyaline, like the epiphyseal plate) and *symphyses* (fibrocartilage, like the pubic symphysis and intervertebral discs).

● **Synovial joints**

Have a joint cavity filled with synovial fluid. Free movement. Most joints of the limbs. This is what RA attacks.

Joints can also be classified *functionally*: **synarthroses** (no motion), **amphiarthroses** (slight motion), **diarthroses** (free motion). Synovial joints are diarthroses; that mobility is exactly what makes them clinically interesting and clinically vulnerable.

Six features that define every synovial joint

Articular (hyaline) cartilage

Caps the ends of the bones. Smooth, glassy, low-friction. No nerves, no blood vessels.

Joint (articular) capsule

Two-layered envelope around the joint. Outer *fibrous layer* (continuous with periosteum) and inner *synovial membrane*.

Synovial membrane

Inner lining of the capsule. Produces synovial fluid.
This is the tissue RA attacks.

Joint cavity

Potential space between the articular cartilages, filled with synovial fluid.

Synovial fluid

Viscous, slippery, hyaluronic-acid rich. Lubricates and nourishes the avascular articular cartilage.

Reinforcing ligaments

Capsular thickenings or separate bands that stabilize the joint and limit unwanted motion.

Where RA strikes. Activated immune cells infiltrate the synovial membrane. The membrane proliferates into an inflammatory mass called *pannus*. Pannus grows over the articular cartilage, releases enzymes that destroy it, and erodes the underlying bone. The classic X-ray finding of *periarticular erosions* is bone being chewed up by pannus where it spreads over the cartilage margins.

Add-ons that protect or improve specific joints

● Menisci

Fibrocartilage pads (knee, TMJ, AC joint). Improve fit, distribute load, absorb shock.

● Labrum

Fibrocartilage rim that deepens a socket (glenoid labrum, acetabular labrum). Improves stability.

● Bursae

Small synovial-fluid-filled sacs where tendons cross bone or ligaments. Reduce friction.
Inflammation = bursitis.

● Tendon sheaths

Elongated bursae wrapping tendons in narrow tunnels (wrist, ankle).

● Fat pads

Cushion the joint where there are spaces (e.g., infrapatellar fat pad).

● Ligaments

Bone-to-bone connectors. Inside the capsule (intracapsular, like ACL), outside it (extracapsular, like MCL), or thickenings of it.

What synovial joints can do, by shape

Synovial-joint movements come in three categories: *gliding* (sliding flat surfaces), *angular* (flexion, extension, abduction, adduction, circumduction), and *rotational* (turning around the long axis). The joint's **shape** determines which motions it allows.

TYPE	SHAPE	MOVEMENT	EXAMPLE
Plane (gliding)	flat-on-flat	short gliding	intercarpal, intertarsal, AC joint
Hinge	cylinder in a trough	flexion/extension only	elbow, knee (mostly), IP joints of fingers
Pivot	rounded end rotating in a ring	rotation around one axis	atlantoaxial (head-shaking "no"), proximal radioulnar
Condylloid (ellipsoidal)	oval convex into oval concave	biaxial: flexion/extension + abduction/adduction	MCP joints (where RA loves to live)
Saddle	both bones saddle-shaped	biaxial, more freedom than condylloid	CMC of thumb (lets thumb oppose)
Ball-and-socket	spherical head in cup	triaxial — full ROM	shoulder (glenohumeral), hip

Factors affecting ROM: shape of articulating surfaces, ligament tightness, muscle bulk, age, prior injury, joint disease. RA reduces ROM by destroying cartilage, stretching ligaments, and triggering muscle spasm.

The two large joints worth knowing in detail

THE SHOULDER (GLENOHUMERAL) JOINT

Ball-and-socket. The most mobile joint in the body, and the least stable. Shallow glenoid receives the much larger humeral head. Stability comes from the **glenoid labrum** (deepens the socket), the **joint capsule** (loose to allow ROM), **glenohumeral ligaments** (capsular thickenings), and the **rotator cuff**: four muscles (supraspinatus, infraspinatus, teres minor, subscapularis — SITS) that surround the joint and actively hold the head in the glenoid.

THE KNEE JOINT

The largest synovial joint in the body. Modified hinge: primarily flexion and extension, with small rotation when flexed. Stability from **cruciate ligaments** (ACL prevents anterior tibial translation, PCL prevents posterior), **collateral ligaments** (MCL medial, LCL lateral), **menisci** (medial and lateral), and the quadriceps tendon-patella-patellar tendon mechanism.

When joints go wrong

● Osteoarthritis (OA)

"Wear and tear." Mechanical loss of articular cartilage over years. Affects weight-bearing joints (hips, knees) and DIPs. NOT immune-mediated.

● Rheumatoid arthritis (RA)

Autoimmune. Synovium attacks the joint. Small symmetric joints. Morning stiffness > 1 hour. Spares DIPs.

● Gout

Monosodium urate crystals deposit in joints. Acute monoarticular, classically big toe MTP (podagra). Sudden, exquisite pain.

● Septic arthritis

Joint infection. Single hot swollen joint with fever. Emergency: requires drainage.

● Bursitis

Inflamed bursa from overuse or trauma. Olecranon, prepatellar, subacromial. Local swelling, pain on pressure.

● Sprain

Stretch or tear of a ligament. Inversion ankle sprain (lateral ligament complex) is the most common.

DRAW ZONE 1

 -- Sketch a generic synovial joint in cross-section. Label articular cartilage, capsule, synovial membrane, joint cavity, synovial fluid, ligaments. Then draw what happens in RA: pannus growing over cartilage.

Use Fine for the synovial folds. Shade the pannus in a different color so the destruction is obvious.



DRAW ZONE 2 -- Draw the six types of synovial joints with an example of each. Then circle the ones RA attacks preferentially (small condyloid + saddle).

MCP and PIP are condyloid; the CMC of the thumb is saddle. RA loves the small ones.

Why these joints, why symmetric, why the DIPs are spared

RA's distribution is anatomical, not random. Once you understand what synovial membrane is and where the largest amount of it lives, the pattern is predictable.

"Morning stiffness > 1 hour"

Inflammatory fluid accumulates in synovial joints overnight when they aren't moving. Movement re-circulates the fluid and disperses inflammatory mediators. RA stiffness lasts > 1 hr; osteoarthritis < 30 min.

"Symmetric distribution"

RA is a **systemic autoimmune disease**. The same antibodies and immune cells circulate to both sides equally, attacking matched synovial joints.

"MCPs, PIPs, wrists, MTPs"

These are small condyloid and saddle synovial joints with a relatively high ratio of synovial membrane to bone. More target tissue per joint = more inflammation per joint.

"DIPs spared"

DIPs have less synovium and a different structural arrangement. They are classically attacked by OA, not RA. Knowing this pattern distinguishes the two at the bedside.

"Periarticular erosions on X-ray"

Pannus (proliferative inflamed synovium) grows over the cartilage margin and into the underlying bone, eroding both. The erosions appear at the joint edges where the synovium starts.

"Why anti-CCP matters"

Anti-cyclic citrullinated peptide antibodies are highly specific for RA, often positive years before symptoms, and predict more aggressive disease. Combined with RF, very supportive of the diagnosis.

Synovium, cartilage, pannus, erosion. Priya's joints are showing you in clinic what a textbook synovial joint diagram never can. When you go back to the prework, the synovial joint cutaway will land differently — it's where her disease lives.

